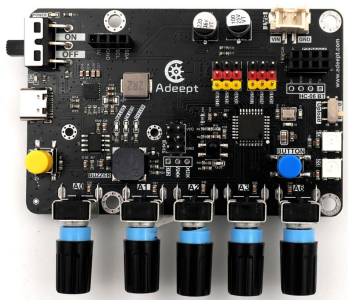


## Lesson 4 Light up the WS2812

### 4.1 Overview

This lesson focuses on how to use the WS2812 LEDs with the Adeept Robotic Arm Control Board. By the end of this lesson, you'll be able to make the WS2812 LEDs display different lighting effects, such as flowing lights, through programming.

### 4.2 Required Components

| Components                       | Quantity | Picture                                                                               |
|----------------------------------|----------|---------------------------------------------------------------------------------------|
| Adeept Robotic Arm Control Board | 1        |  |
| Type-C USB Cable                 | 1        |  |

### 4.3 Principle Introduction

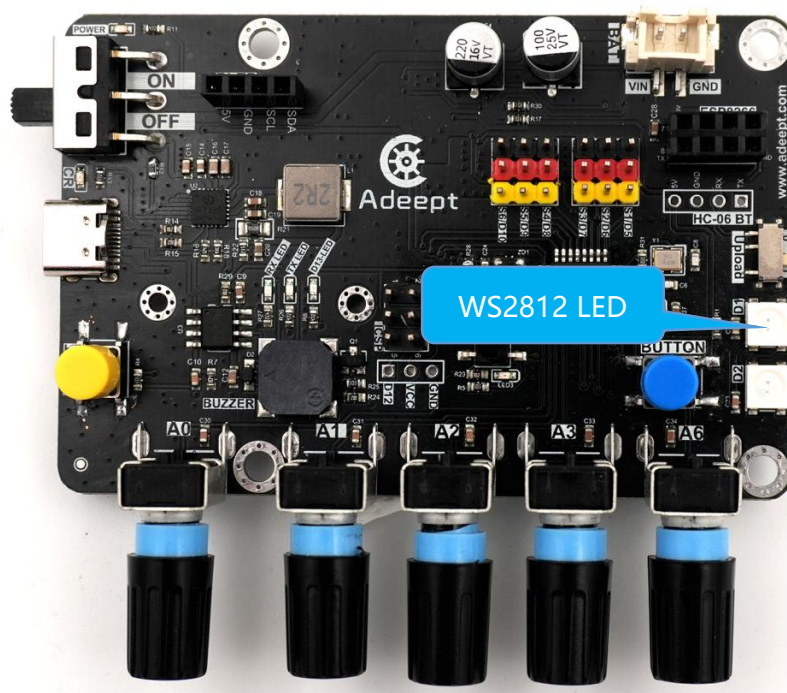
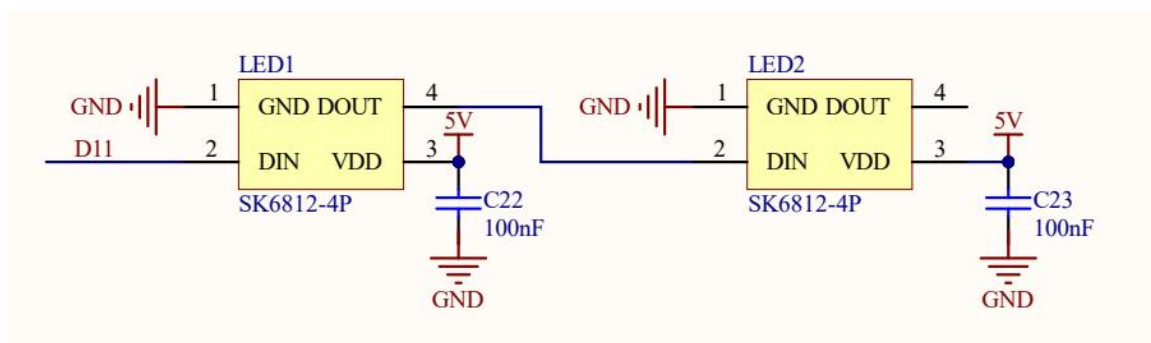
Red, green, and blue are called the three primary colors. When you combine these three primary colors of different brightness, it can produce almost all kinds of visible light.

WS2812 RGB module is a low-power RGB tri-color lamp with integrated current control chip. Its appearance is the same as a 5050LED lamp bead, and each element is a pixel. The pixel contains an intelligent digital interface data latch signal shaping amplifier driving circuit, and also contains

a high-precision internal oscillator and a 12V high-voltage programmable constant current control part, which effectively guarantees that the color of the pixel light is highly consistent.

Two WS2812 LEDs are integrated on the Adept Robotic Arm Control Board.

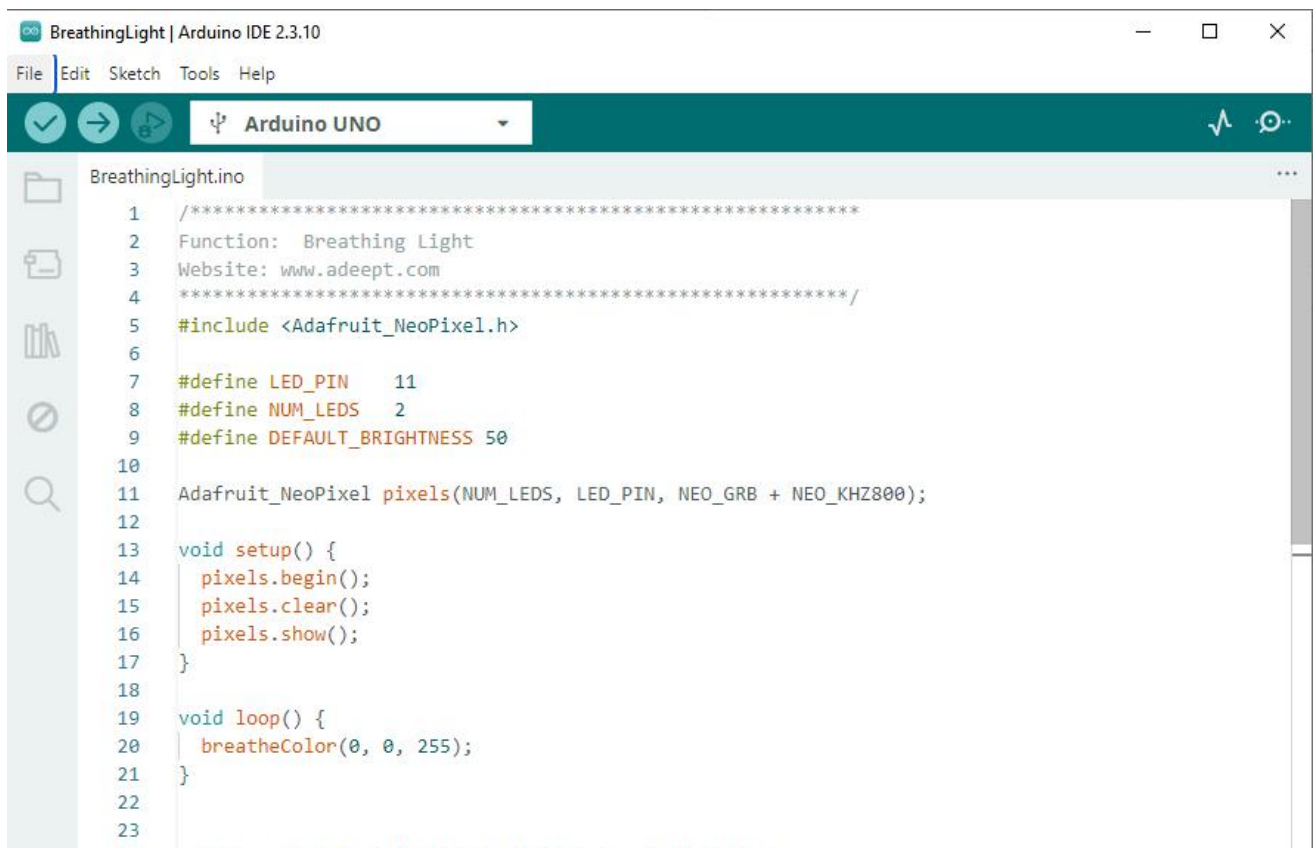
## 4.4 Wiring Diagram



| PINS of Adeept Robotic Arm Control Board | WS2812 |
|------------------------------------------|--------|
| D11                                      | WS2812 |

## 4.5 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board (Arduino Board) with a USB cable.
2. Open "**04\_WS2812/BreathingLight**" folder in **"/Code"** , double-click "**BreathingLight.ino**" .



```

BreathingLight.ino
1  /*****
2  Function: Breathing Light
3  Website: www.adeept.com
4  *****/
5  #include <Adafruit_NeoPixel.h>
6
7  #define LED_PIN    11
8  #define NUM_LEDS   2
9  #define DEFAULT_BRIGHTNESS 50
10
11  Adafruit_NeoPixel pixels(NUM_LEDS, LED_PIN, NEO_GRB + NEO_KHZ800);
12
13  void setup() {
14    pixels.begin();
15    pixels.clear();
16    pixels.show();
17  }
18
19  void loop() {
20    breatheColor(0, 0, 255);
21  }
22
23

```

3. Select development board and serial port.

Board: **Tools---**>**Board---**>**Arduino AVR Boards---**>**Arduino Uno**

Port: **Tools --->Port---**>**COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```

Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.

```

5. After the program runs successfully, the WS2812 LEDs will show a breathing light effect. The brightness of the light will slowly alternate between dim and bright.

## 4.6 Code

```

01  /*****
02  Function: Breathing Light
03  Website: www.adeept.com
04  *****/
05  #include <Adafruit_NeoPixel.h>
06
07  #define LED_PIN    11
08  #define NUM_LEDS   2
09  #define DEFAULT_BRIGHTNESS 50
10
11  Adafruit_NeoPixel pixels(NUM_LEDS, LED_PIN, NEO_GRB + NEO_KHZ800);
12
13  void setup() {
14    pixels.begin();
15    pixels.clear();
16    pixels.show();
17  }
18
19  void loop() {
20    breatheColor(0, 0, 255);
21  }
22
23
24  void breatheColor(uint8_t R, uint8_t G, uint8_t B) {
25    for (int i = 0; i <= 255; i++) {
26      setAllLED(R, G, B, i);
27      delay(8);
28    }
29    for (int i = 255; i >= 0; i--) {
30      setAllLED(R, G, B, i);
31      delay(8);

```

```
32     }
33 }
34
35 void setAllLED(uint8_t R, uint8_t G, uint8_t B, uint8_t brightness) {
36     uint8_t r = map(brightness, 0, 255, 0, R);
37     uint8_t g = map(brightness, 0, 255, 0, G);
38     uint8_t b = map(brightness, 0, 255, 0, B);
39
40     for (int i = 0; i < NUM_LEDS; i++) {
41         pixels.setPixelColor(i, pixels.Color(r, g, b));
42     }
43     pixels.show();
44 }
```